

Correction.

Problem

- **Input:** A sequence of US coins: penny (1¢), nickel (5¢), dime (10¢), quarter (25¢).
- **Goal:** Accept the input **only if the first few coins add up to exactly 30¢**.
- **Output:** Acknowledgement (accepting state) when 30¢ is reached.

Define the States

Each state represents the **total value accumulated so far**.

- **Start State:** 0¢
- **Intermediate States:** 1¢, 2¢, ..., 29¢
- **Accepting State:** 30¢
- **Rejecting States:** Any state where the total exceeds 30¢

So, we'll have **31 states** (from 0 to 30), plus an optional **reject state** for totals >30.

Define the Alphabet

The input symbols (alphabet) are the coin values:

- $\Sigma = \{1, 5, 10, 25\}$

Define Transitions

From each state, define transitions based on adding a coin:

- From state **s**, on input **c**, go to state **s + c** if **s + c ≤ 30**
- If **s + c > 30**, go to **reject state**

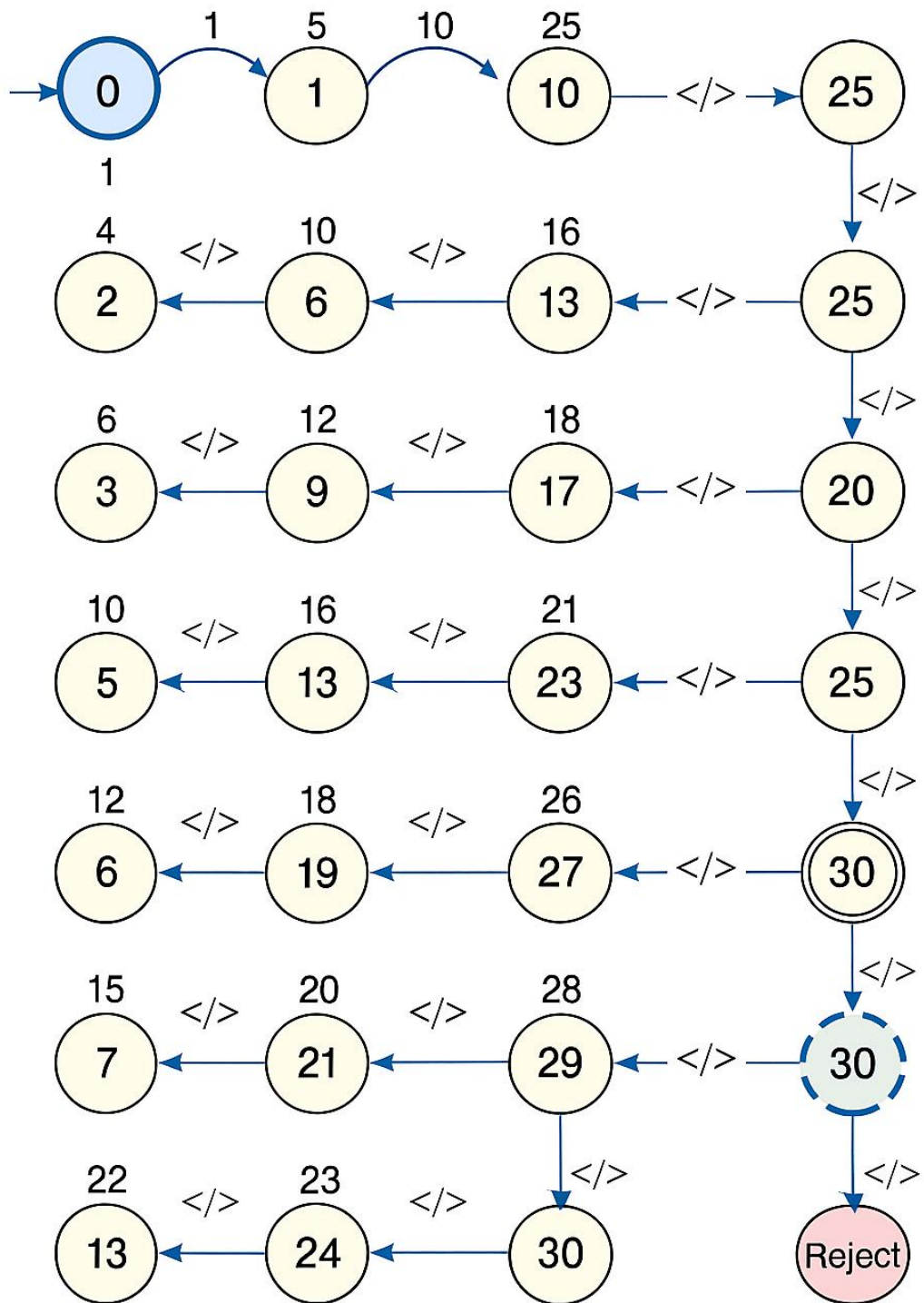
Example:

- From state 0:
 - on 1 $\rightarrow$ state 1
 - on 5 $\rightarrow$ state 5
 - on 10 $\rightarrow$ state 10
 - on 25 $\rightarrow$ state 25
- From state 25:
 - on 5 $\rightarrow$ state 30 (accepting)
 - on 1/10/25 $\rightarrow$ reject

Define Accepting and Rejecting Conditions

- **Accepting State:** Only state 30
- **Rejecting State:** Any state where total >30

Draw the FSA



FSA Explanation for 30¢ Coin Recognition

Goal

Design an FSA that:

- Accepts US coins: **1¢ (penny), 5¢ (nickel), 10¢ (dime), 25¢ (quarter)**
- Acknowledges when the **first coins entered** sum to **exactly 30¢**

Core Idea

Each **state** represents the **total value** of coins entered so far. Transitions between states occur when a coin is added. The FSA **accepts** when the total reaches **exactly 30¢**.

Alphabet (Input Symbols)

- $\Sigma = \{1, 5, 10, 25\}$

States

- **Start State:** 0¢
- **Intermediate States:** 1¢, 2¢, ..., 29¢
- **Accepting State:** 30¢
- **Reject State:** Any state where total >30¢

Transitions

From each state, transitions are defined by adding a coin:

- From state s , on input c , go to state $s + c$ if $s + c \leq 30$
- If $s + c > 30$, go to a **reject state**

Example Transitions:

- From state 0:
 - on 1 $\rightarrow$ state 1
 - on 5 $\rightarrow$ state 5
 - on 10 $\rightarrow$ state 10
 - on 25 $\rightarrow$ state 25
- From state 25:
 - on 5 $\rightarrow$ state 30 (accepting)
 - on 1/10/25 $\rightarrow$ reject

Accepting State

- **State 30:** This is the only accepting state. If the first few coins sum to 30¢, the FSA enters this state and **accepts** the input.

Rejecting State

- Any transition that causes the total to exceed 30¢ leads to a **reject state**. Once in this state, the FSA does **not accept** the input.

Diagram Notes

- Each node is labeled with the current total (e.g., 0, 5, 10, ..., 30)
- Arrows between nodes are labeled with coin values (1, 5, 10, 25)
- The accepting state (30) is marked with a **double circle**
- The reject state is for clarity

E#3 - Module 3: In-Class Exercises #3

Formal Definition of the FSA

A Finite State Automaton is defined as a 5-tuple:

$$M = (Q, \Sigma, \delta, q_0, F)$$

- **Q (States):**

$$Q = \{ q_0, q_1, q_5, q_{10}, q_{15}, q_{20}, q_{25}, q_{30} \}$$

(Complete set for pennies): $Q = \{ q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{10}, q_{11}, q_{12}, q_{13}, q_{14}, q_{15}, q_{16}, q_{17}, q_{18}, q_{19}, q_{20}, q_{21}, q_{22}, q_{23}, q_{24}, q_{25}, q_{26}, q_{27}, q_{28}, q_{29}, q_{30} \}$

(Complete set for pennies):

$$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{10}, q_{11}, q_{12}, q_{13}, q_{14}, q_{15}, q_{16}, q_{17}, q_{18}, q_{19}, q_{20}, q_{21}, q_{22}, q_{23}, q_{24}, q_{25}, q_{26}, q_{27}, q_{28}, q_{29}, q_{30} \}$$

(Optional: add q_{dead} for overshoot handling)

All intermediate states for pennies (q_2, q_3, q_4 , etc.).

Each state represents the cumulative amount entered so far.

- **Σ (Alphabet):**

$$\Sigma = \{ 1, 5, 10, 25 \}$$

Where:

- 1 = penny
- 5 = nickel
- 10 = dime
- 25 = quarter

- **q_0 (Initial State):**

$$q_0 \text{ (represents 0¢)}$$

- **F (Final States):**

$$F = \{ q_{30} \}$$

(Note: $q_{dead} \notin F$)

- $\$$ (acknowledgement state when 30¢ is reached)

- **δ (Transition Function):**

For each state, add the coin value if the sum ≤ 30 ¢.

Examples:

- $\delta(q_0, 5) = q_5$
- $\delta(q_0, 10) = q_{10}$
- $\delta(q_5, 25) = q_{30}$
- $\delta(q_{10}, 10) = q_{20}$
- $\delta(q_{25}, 5) = q_{30}$

Full rule:

For all $i \in \{0..30\}$ and coin $c \in \Sigma = \{1,5,10,25\}$:

If $i + c \leq 30$: $\delta(q_i, c) = q_{i+c}$

If $i + c > 30$: $\delta(q_i, c) = q_{dead}$ (optional for DFA completeness)

For q_{30} (accepting sink): $\delta(q_{30}, c) = q_{30}$ for all $c \in \Sigma$

For q_{dead} (if used): $\delta(q_{dead}, c) = q_{dead}$

If adding a coin exceeds 30¢, ignore or stay in q_{30} .

Full rule:

For all $i \in \{0..30\}$ and coin $c \in \Sigma = \{1,5,10,25\}$:

If $i + c \leq 30$: $\delta(q_i, c) = q_{i+c}$

If $i + c > 30$: $\delta(q_i, c) = q_{dead}$ (optional for DFA completeness)

For q_{30} (accepting sink): $\delta(q_{30}, c) = q_{30}$ for all $c \in \Sigma$

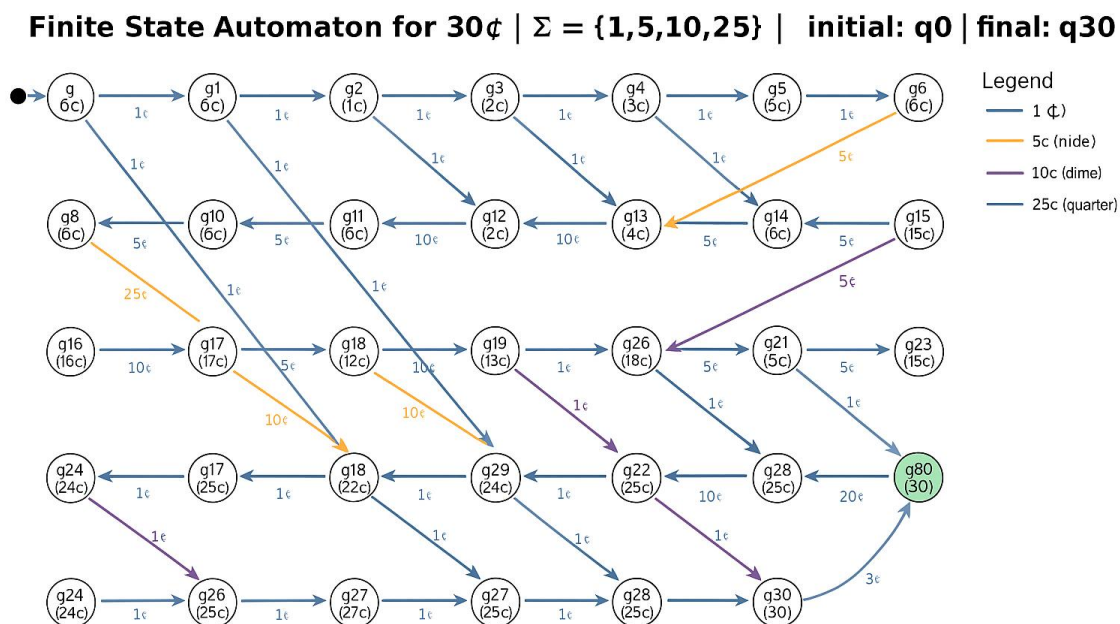
For q_{dead} (if used): $\delta(q_{dead}, c) = q_{dead}$

Complete Transition Function (δ)

Diagram Structure

- **Nodes:** $q_0 \rightarrow q_1 \rightarrow q_5 \rightarrow q_{10} \rightarrow q_{15} \rightarrow q_{20} \rightarrow q_{25} \rightarrow q_{30}$
- **Edges:** Labeled with coin values (1, 5, 10, 25)
- **Final State:** q_{30} (double circle)
- **Initial State:** q_0 (arrow pointing to it)

The diagram should include all states $q_0 \dots q_{30}$ and show q_{30} as a double circle with a self-loop for Σ .



Example Transition Table

Current State Input Next State

q_0 1 q_1

q_0 5 q_5

Current State Input Next State

q_0 10 q_{10}

q_0 25 q_{25}

q_5 5 q_{10}

q_5 10 q_{15}

q_5 25 q_{30}

...